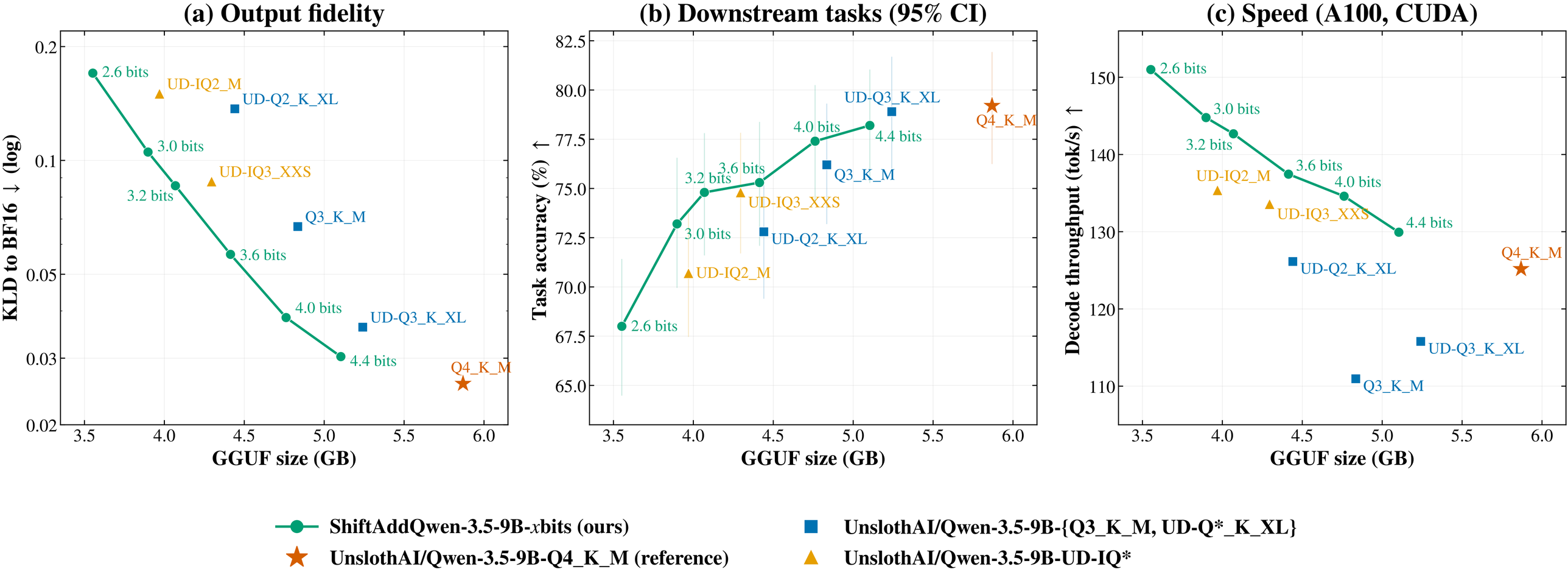
(a) Output fidelity
KLD to BF16 ↓ (log)
GGUF size (GB)
2.6 bits
3.0 bits
3.2 bits
3.6 bits
4.0 bits
4.4 bits
UD-IQ2_M
UD-Q2_K_XL
UD-IQ3_XXS
Q3_K_M
UD-Q3_K_XL
Q4_K_M
(b) Downstream tasks (95% CI)
Task accuracy (%) ↑
GGUF size (GB)
(c) Speed (A100, CUDA)
Decode throughput (tok/s) ↑
GGUF size (GB)
ShiftAddQwen-3.5-9B-xbits (ours)
UnslothAI/Qwen-3.5-9B-Q4_K_M (reference)
UnslothAI/Qwen-3.5-9B-{Q3_K_M, UD-Q*_K_XL}
UnslothAI/Qwen-3.5-9B-UD-IQ*